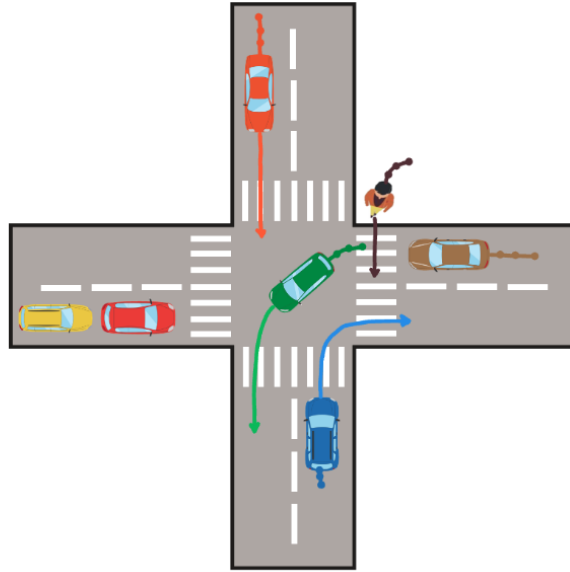




MILESTONE 3
CIVIL-459 : DEEP LEARNING FOR AUTONOMOUS VEHICLES

PROFESSOR : ALEXANDRE ALAHI

Trajectory prediction

Group : The bests
Alexandre Vray 310104
Loïc Finette 296932

Spring 2024

1 Introduction

In this last Milestone, the goal is to implement a new model in the Unitraj framework. The model we tried to implement is called GameFormer and comes from [1]. This model should be able to outperform the state of the art models with 0.9161 minADE on WOMB dataset.

2 Gameformer

The Gameformer model is based on game theory. It makes predictions for many agents, and then tries to improve on those predictions using predicted interactions. This process is repeated on a K level decoder as illustrate in Figure 1a. The model is hence composed in three main parts, the encoder, the multilevel decoder and the best level prediction.

The encoder proposed in [1] is not compatible with our dataset since it requires the map lines from every agents point of view but we only have one overall map lines for the scene. Hence we decided to use the same encoder as proposed in the ptr model and encode the scene but not only from ego but from every agents point of view.

The decoder is mainly the same as proposed in [1] and illustrate in Figure 1b. There is first an initial stage decoder which predicts trajectories only based on past features. Then the k -level decoder adapts the predictions based on interactions.

The scores computed at each level should allow us to select the best level prediction as the model output.

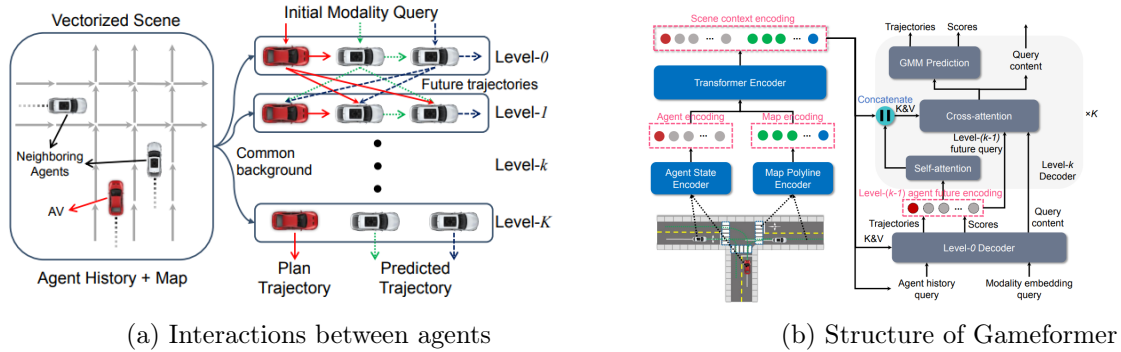


Figure 1: Gamformer illustrations

3 Code

Please find our submission code [here](#). A new model gameformer has been added to the framework, please pay attention on the ReadMe file for detailed explanations of the code.

4 Performance

We want to compare our new gameformer model with the previous ptr model. In the plots and tables, the ptr baseline is the first ptr model proposed in Milestone 1. Ptr best is the improved ptr model

ptroposed in Milestone 2. Gameformer is the actual model with score based best level selection and gameformer ideal is what we would expect if the model is effectively able to well fit the scores. (Not the case in our actual implementation :-).

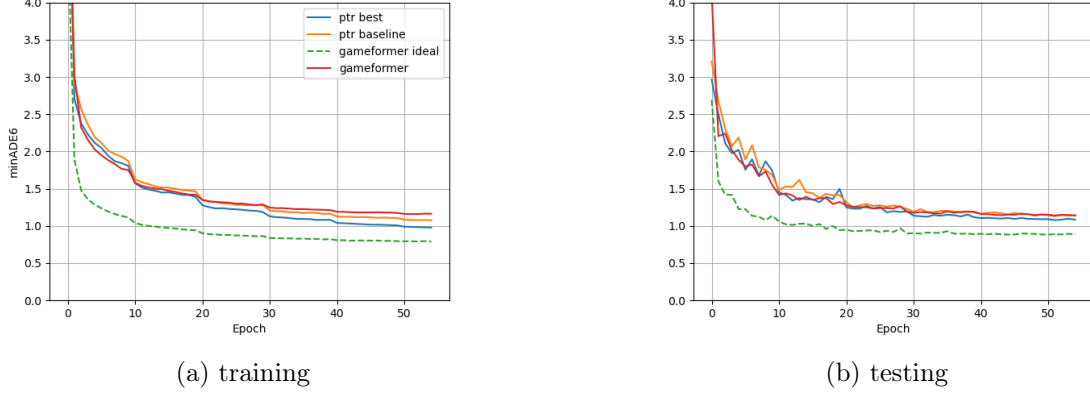


Figure 2: minADE6 training evolution

From the above graphs (Figure 2), we first observe the gameformer decoder does not fit the train data as well as the ptr models. We also observe less overfitting with our new model since the results are competitive on the validation set.

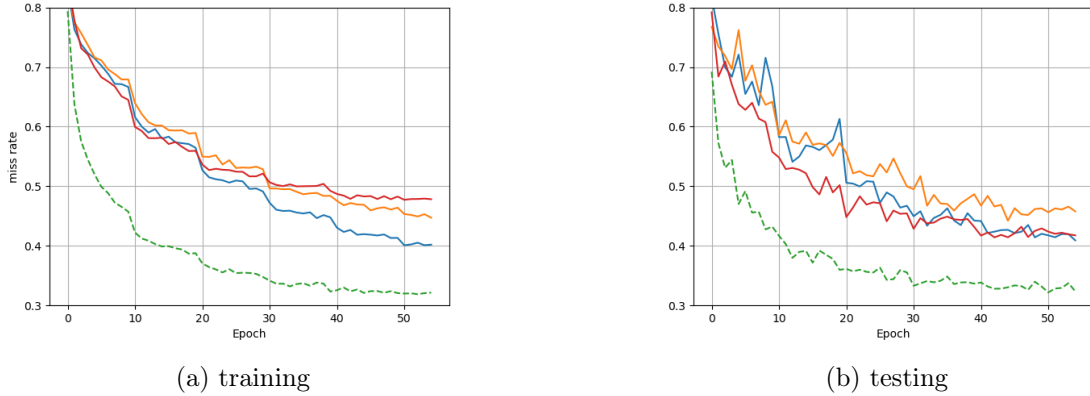


Figure 3: miss rate training evolution

In Figure 3b, we observe that the gameformer model is competitive with the ptr decoder in terms of missrate. Our best ptr performed better on minADE6 score. Maybe the interaction decoder has a bigger impact on reducing the miss rates than the minADE6.

	Training set			Validation set			Kaggle
	minADE6	minFDE6	miss rate	minADE6	minFDE6	miss rate	minADE6
baseline ptr	0.940	2.055	0.392	1.124	2.508	0.439	...
best ptr	0.920	2.000	0.371	1.087	2.435	0.408	0.989
level 0	...	...	...	1.194	2.788	0.462	...
level 1	...	...	...	1.140	2.600	0.422	...
level 2	...	...	...	1.139	2.603	0.418	1.081
level 3	...	...	...	1.130	2.611	0.424	1.074
level 4	...	...	...	1.163	2.708	0.431	...
level selection	1.164	2.655	0.478	1.141	2.603	0.418	1.081
ideal gameformer	0.790	1.912	0.316	0.894	2.112	0.328	...

Table 1: Metrics overview on training and validation sets

In Table 1 we summarize the metrics scores of our models. We also compare the scores with every output level of the **gameformer** model. The difference in overfitting is really highlighted between **best ptr** model and the **gameformer** ones. It is also clear that with a good score prediction, the **gameformer** model has a huge potential that we were not able to implement in this short time-slot project.

5 Future work

- Build a more suitable loss function to better predict best level / mode.
- Use a better initial level predictor (as ptr) and then perform interactions from it.
- Build upon our improvements by adding other improvements such as handling long tailed distributions.

6 Conclusion

From this implementation, we can conclude that the **gameformer** model is competitive with the **ptr** one. It displays less overfitting during training process but actually does not improve the **ptr** predictions. But the potential of this model is huge if we are able to properly fit the scores prediction with the ground truth minADE6.

References

- [1] Z. Huang, H. Liu, and C. Lv, “Gameformer : Game-theoretic modeling and learning of transformer-based interactive prediction and planning for autonomous driving,” 2023.